

Abichal Ghosh

480-876-3640 | aghosh55@asu.edu | [linkedin.com/in/abichal-ghosh/](https://www.linkedin.com/in/abichal-ghosh/) | github.com/ABICHAL1708

EDUCATION

Arizona State University

Master of Science, Computer Science

Aug. 2022 – May 2024

GPA - 4.00/4.00

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering, Computer Science, First Division

Aug. 2018 – May 2022

GPA - 8.65/10.00

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, Java, MATLAB, HTML 5, CSS, Bash, MySQL

Tools: D3, React, Git, AWS, DynamoDB, TensorFlow, PyTorch, NumPy, SciPy, scikit-learn, Pandas, Linux, FastAPI

Subjects: Cloud Computing, Software Design, Data Structures & Algorithms, Data Mining, Natural Language Processing, Semantic Web Mining, Information Assurance & Security, Data visualization, Human-Computer Interaction, Computer Vision, Artificial Intelligence, Web Development, Database Management Systems, Operating Systems

EXPERIENCE

Machine Learning Engineer

501CThree.Corp

July 2024 – Current

California, USA

- Detected anomalies for enabling accurate failure detection in the data collected from the Raspberry Pis.
- Trained a classification model with 85.7% accuracy on data with effective interpolation of missing values.
- Designed and trained an LSTM based model with 92% accuracy for the classification of Time-Series data.

Software Developer Engineer

501CThree.Corp

May 2023 – May 2024

Michigan, USA

- Refactored and debugged existing code for data collection from Raspberry Pis resulting in improved efficiency.
- Implemented a new streamlined distributed system with separate folders and AWS DynamoDB tables to enable simultaneous real-time data collection and upload from multiple Raspberry Pis across 4 different locations.
- Utilized AWS CloudWatch, Lambda, SQS, and SNS to create a real-time alarm system for the Raspberry Pis.

Machine Learning Researcher

Computer Vision Center (CVC, UAB, Spain)

May 2021 – Dec 2021

Barcelona, Spain

- Analyzed image compression codecs and Variational Autoencoder(VAE) models to identify areas for improvement.
- Visualized the Filters of the 1st layer of the models, and examined subsequent layers of the models using PCA, CKA, Canonical Correlational Analysis (CCA), Singular Vector Canonical Correlational Analysis (SVCCA), etc.
- Improved the size and the performance by evaluating a prototype, achieving an average improvement of 47.5%.

Machine Learning Intern

RRSC West, NRSC, ISRO

May 2020 – June 2020

Rajasthan, India

- Generated and exported 500+ Satellite Images from Google Earth Engine using JavaScript & Data Augmentation.
- Utilized DeepLabv3+ and UNet for high-res Satellite Image Segmentation to detect Solar Farms and Sand Dunes.
- Achieved autonomous target detection for the precise tracking and monitoring of solar farms(for maintenance) and sand dunes(An indicator for managing disasters, particularly sandstorms).

PROJECTS

Facial Image Recognition | AWS EC2, AWS Lambda, Python, PyTorch, FastAPI

Jan 2024 – May 2024

- Deployed a facial image recognition application using Cloud, utilizing AWS S3, SQS, EC2, and Lambda, ensuring the adherence to software engineering best practices, including design patterns, Git, and Docker
- Achieved high concurrency with Lambda, processing videos into frames with more than 5 concurrent instances and an average duration of less than 8 seconds per invocation.
- Implemented auto-scaling with SQS and EC2, scaling to a maximum of 20 EC2 instances based on the workload, achieving sub-3-second response times for 1000 concurrent requests.

Effect of Honest Examples in LLM | Python, LLMs, PEFT, Finetuning

Aug 2023 – Dec 2023

- Generated a diverse dataset consisting of 800 'unsafe' questions in the format of the Alpaca dataset.

- Finetuned the Llama 7-billion model using PEFT-LORA on subsets of the curated dataset to evaluate the effect of incorporating 'unsafe' questions on the model's ethical and safety consideration.
- Evaluated the generalizability and conservativeness of the finetuned models on "Do not answer" and "LLM attacks" dataset with the highest generalizability accuracy of 92.27% and highest conservativeness accuracy of 90%.

Prototyping and Redesigning a website | *Javascript, React, HCI, UI/UX*

Jan 2023 – May 2023

- Led a Human-Computer Interaction project focused on redesigning a website employing React for the redesign.
- Enhanced user experience by introducing a cart feature and resolving critical page linking errors.
- Evaluated project success with metrics including 53.27% increase in Overall Relative Efficiency, 27.8% decrease in average navigation time and 65.41% increase in the mean System Usability Score.

Effective Image Watermarking | *Python, Pytorch, Computer Vision, GANs*

Jan 2023 – May 2023

- Curated a diverse dataset of 12779 watermarked images comprising of 6 different watermark variations.
- Utilized YOLO v5 and U2Net for precise watermark detection and GANs for the successful watermark removal.
- Assessed watermark robustness via SSIM, with the top variation exhibiting a 16.4% increase over the basic version

Cloud System Visualization with Behavioral Lines | *Javascript, D3.js, AWS*

Aug 2022 – Dec 2022

- Leveraged a dataset with 1440 data points, containing 5 performance metrics of multiple clusters of AWS servers.
- Implemented an interactive and flexible visualization using D3.js consisting of 4 charts to analyze the highly complex and correlated cloud computing dataset.

UML Diagram Generator | *Java, Software Design, Software Architecture*

Aug 2022 – Dec 2022

- Developed a dynamic UML diagram generator using Java, showcasing proficiency in software design & architecture
- Designed a GUI with a dual-interface system: one side for code structure input, and the other for the interactive UML visualization enabling the users to intuitively add class relations by interacting with the diagram elements.
- Utilized the Model-View-Controller(MVC) design architecture pattern for a more modular architecture.
- Employed industry-standard practices such as Decorator Design Pattern for dynamically adding GUI elements in real-time and Observer Design Pattern for real-time code-diagram synchronization.

PUBLICATIONS

-
- [1] **Ghosh, A.**, et al. 2023. Industrial Revolution 4.0 With a Focus on Food-Energy-Water Sectors. In Encyclopedia of Data Science and Machine Learning (pp. 2199-2210). J. Wang, Ed. IGI Global. DOI: 10.4018/978-1-7998-9220-5.ch131